# Simplex Takehome

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## 1 Non-ergodic Mess3 data setup

We construct a dataset with 3 distinct Mess3 processes. Let us refer to these 3 processes as the K=3 components of our setup. We build our dataset by sampling uniformly from these 3 components. That is $\mathbb{P}(\text{in component k}) = w_k = \frac{1}{K} = \frac{1}{3} \quad \forall k \in \{1, 2, 3\}$. We define the parameters of our three Mess3 proceses as below:

- Component 0: $(\alpha = 0.9, x = 0.1)$
- Component 1: $(\alpha = 0.7, x = 0.2)$
- Component 2: $(\alpha = 0.5, x = 0.3)$

The idea here is to make the data generating processes relatively distinct. Component 0 has highly informative emission and persistent states, Component 2 has more uninformative emissions with frequent state transitions, and Component 1 is somewhere in between. The mixing eigenvalues $(1 - 3x)$ are 0.7, 0.4, and 0.1 respectively, which control how quickly beliefs converge and the fractal detail of each component's mixed-state presentation.

This setup is interesting and relevant to language models because language is non-ergodic. Language models must learn to 1. identify what data generating process it is modeling (e.g., is this text from an angry redditor, political speech, french person, etc which maps to selecting which component is generating this data) and 2. learn what that data generating process is (e.g., how does a redditor speak here, how does a political speak, etc which maps to learning the specific mess3 process). Insights from this setup can inform big topics around how models learn to use different data generating processes, like the persona selection model or in context learning.

**Training setup.** We train a 1-layer GPT-2-style (pre-norm) transformer with $d_{\text{model}} = 64$, 4 attention heads, $d_{\text{mlp}} = 256$, GELU activations, and no dropout, totaling 51,264 parameters. The model uses causal self-attention over a context window of 12 tokens with vocabulary size 3. We train with Adam (lr $= 10^{-3}$) for 25,000 steps using online data generation: each batch of 64 sequences is freshly sampled, so the model never sees the same sequence twice. The loss is cross-entropy on next-token prediction at all 12 positions simultaneously.

## 2 Activation Geometry Prediction

In the non-ergodic setup, at each token position, the model must encode a joint belief state that decomposes into two parts: which Mess3 component is active, and what hidden state that component is in. Unlike

in the "Transformers learn factored representations" paper, where all $N$ factors are simultaneously active (tensor product transitions), only one Mess3 process is active per sequence. This gives a block-diagonal joint transition matrix $T^{(z)}_{\text{joint}} = \text{diag}(T^{(z)}_1, \ldots, T^{(z)}_K)$, so the belief state is a mixture $(w_1\eta_1, \ldots, w_K\eta_K)$ instead of a than a tensor product. The full belief lives on a $(3K-1)$-simplex, constrained to the convex hull of the $K$ component-specific 2-simplices.

Starting at position 1, all three components have uniform stationary distributions, so a single token gives zero information about the component. Differentiation should begin at position 2, where we start to gain information on the transition dynamics. This means effective dimensionality should increase with token position: early positions have few distinct belief states (3 at position 1, 9 at position 2), while later positions should spread beliefs across more of the $(3K-1)$-simplex. Dimensionality would only decrease at long contexts where component posteriors become one-hot (leaving us with one block from our joint transition matrix), but for context length 12, the model should still carry substantial component uncertainty.

Conditioned on a single component, the activation geometry should look like (from Piotrowski et al. "Constrained Belief Updates Explain Geometric Structures in Transformer Representations"):

- The attention layer writes a coarser, constrained approximation of the belief fractal
- The MLP refines toward the full MSP fractal.

This implies that at the positions the model becomes confident about which component it is in, the total activation geometry should look like K=3 distinct 2D fractals, where each fractal evolves as above across layers.

For the overall geometry, I see three main possibilities:

1. **Superimposed fractals separating with context.** Early on, the three components' geometries overlap in a shared subspace. As context grows, they drift apart into distinct clusters with internal fractal structure.
2. **Component ID and belief in separate subspaces.** A few dimensions encode the posterior $w_k$, while a shared 2D subspace encodes the within-component belief.
3. $K$ **orthogonal 2D subspaces.** Each component gets its own subspace, with sparsity encoding component identity.

My primary prediction is (1), since it follows most naturally from Bayesian updating in the joint space, and I'd imagine the model must maintain a blended representation for much of the context window.

# 3 Training Results

The model converges close to the Bayes-optimal cross-entropy, computed via Monte Carlo estimation (25,000 samples) of the full Bayesian predictor that maintains posterior beliefs over all three components and all hidden states jointly. The mean Bayes-optimal loss across positions is 1.048 nats.

**Additional analysis 1.** We also decompose the loss by component, comparing the model's per-component CE to the Bayes-optimal oracle that knows which component generated the sequence. This decomposition is informative because the three components have very different mixing eigenvalues $(1-3x) \in 0.7, 0.4, 0.1$, which govern how quickly belief states converge and therefore how much each component benefits from additional context. By separating the loss, we can see whether the model is uniformly close to optimal or whether it struggles more with certain components.

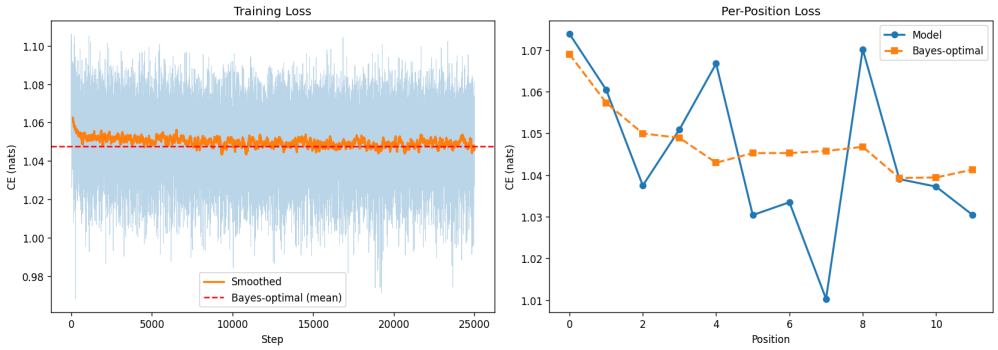


Figure 1: Left: training cross-entropy with Bayes-optimal mean (dashed red). Right: per-position CE comparing the trained model (circles) to the Bayes-optimal predictor (squares).

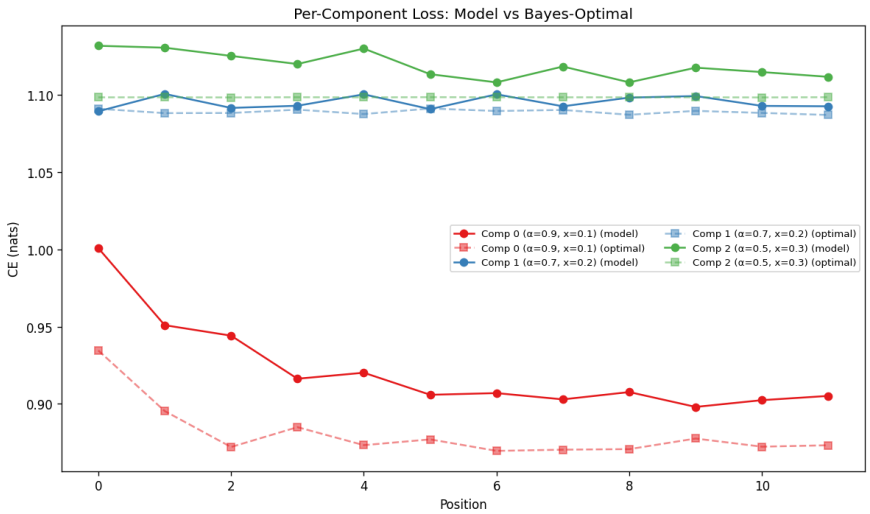


Figure 2: Per-component cross-entropy: model (solid) vs Bayes-optimal oracle (dashed) for each component.

Component 0 ($x = 0.1$, eigenvalue 0.7) shows the largest gap between early and late positions: additional context substantially improves prediction because states are persistent and emissions are informative. The model tracks the oracle curve closely, staying consistently just above optimal. Components 1 and 2 are essentially flat after position $\sim$2–3. With eigenvalues 0.4 and 0.1, their mixing is fast enough that the belief state is close to converged after just a few tokens. The CE plateaus at a level set by the stationary prediction quality, and the model matches this plateau well.

The key takeaway is that the model's suboptimality is roughly uniform across components rather than concentrated on any one. It doesn't sacrifice performance on hard-to-predict components (Component 2) to do better on easy ones (Component 0). This suggests the model has learned a more general-purpose belief tracking mechanism rather than specializing for a subset of the mixture.

# 4 Residual Stream Geometry

## 4.1 Effective Dimensionality

We compute PCA of the post-MLP activations at each of the 12 positions and measure effective dimensionality as the number of principal components needed to explain 95% of variance. The effective dimensionality rises from 2 at position 0 to 5 at positions 6–11, where it plateaus.

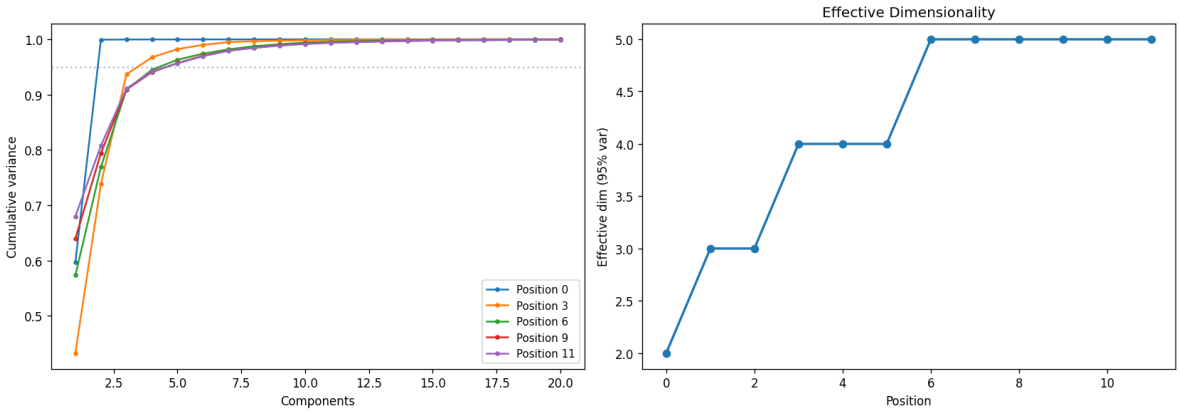


Figure 3: Left: cumulative PCA variance at selected positions. Right: effective dimensionality (95% variance threshold) vs context position.

This fits with our later analysis which indicates that the model uses a shared subspace for within-component belief and separate dimensions for encoding the component posterior $w_k$, totaling fewer effective dimensions than the full joint belief would require. The effective dimensionality of 5 is consistent with a near-factored representation (close to the 4-dimensional lower bound of a shared 2D belief subspace plus a 2D component posterior), and well below the 8 dimensions required by the full joint belief. The extra dimension likely captures interactions between component identity and belief dynamics, or residual token-level information. This would be a good question to explore more if I had more time.

## 4.2 Linear Decodability of Beliefs

We fit Ridge regression ($\alpha = 10$) from post-MLP activations to three belief targets using an 80/20 train/test split: the joint belief (9D, $[w_0\eta_0, w_1\eta_1, w_2\eta_2]$), within-component belief (3D, $\eta_k$ using the true component), and component posterior (3D, $[w_0, w_1, w_2]$). At the last position, the test $R^2$ values are:

- Joint belief: $R^2 = 0.835$
- Within-component belief: $R^2 = 0.819$
- Component posterior: $R^2 = 0.802$

All three targets show $R^2$ increasing with position, starting near zero at position 0 (where beliefs are uninformative).

All three belief representations are linearly decodable from the residual stream with $R^2 > 0.8$ at the last position, supporting that the model encodes Bayesian belief states as linear features.

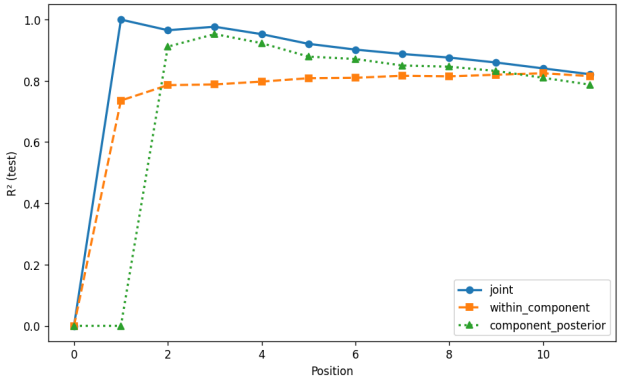


Figure 4: Test $R^2$ of Ridge regression from post-MLP activations to belief representations at each context position.

The three curves have distinct trajectories. The within-component belief jumps at position 1 and stays roughly flat, so a single token substantially updates the hidden state belief, and additional tokens seem to refine it only marginally. The component posterior is zero at position 1, since all components share the same uniform stationary distribution and a single token cannot distinguish them. It jumps at position 2 when transition dynamics first become visible, matching our prediction from Section 2. The joint belief peaks at position 1 and slightly decreases thereafter: early on, the uniform component posterior makes the 9D joint target effectively lower-dimensional and easy to fit linearly, but as the posterior differentiates with context, what seems to be happening is that the joint becomes a higher-dimensional target that is harder for a single linear map to capture.

## 4.3 Component Separation in PCA

We visualize 2D PCA of post-MLP activations at positions 0, 2, 5, 8, and 11, colored by component identity. At early positions the three components completely overlap. By later positions, visible clusters emerge, though the separation is not clean in 2D. In 3D PCA (capturing 91.1% of variance at position 11), the component clusters are not much more distinguishable.

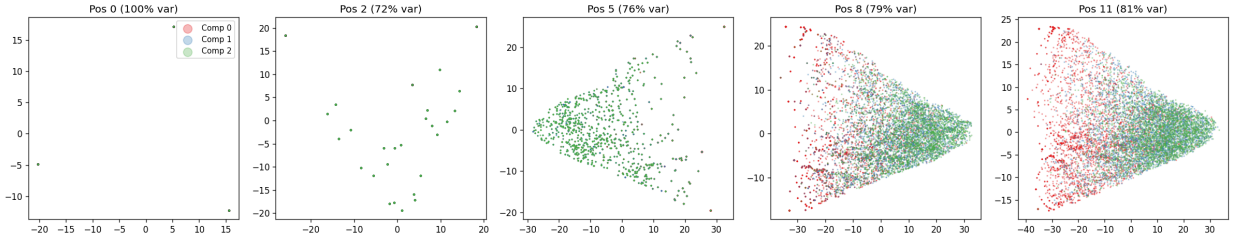


Figure 5: 2D PCA of post-MLP activations colored by component identity at selected positions.

The lack of clean separation in the top 2–3 PCs is itself informative. PCA finds the directions of maximum variance, which here are dominated by within-component belief dynamics. Component identity is encoded in the residual stream (the component posterior is linearly decodable with $R^2 = 0.802$), but in dimensions that explain less variance and therefore don't appear in the top PCs. This is consistent with the shared-subspace finding from Section 4.5: the high-variance belief subspace is shared across components, while component identity lives in quieter directions orthogonal to it.

## 4.4 Fractal Structure

We project the residual stream onto the 2-simplex via a learned linear map: for each component, we fit Ridge regression from activations to within-component beliefs, then project the predictions onto the simplex. Colors represent the true within-component belief state (RGB = probability of each hidden state).

Component 0 (eigenvalue 0.7) shows clear fractal structure at late positions, consistent with the mixed-state presentation predicted by theory. Component 1 (eigenvalue 0.4) shows moderate structure. Component 2 (eigenvalue 0.1) collapses to a small blob near the simplex center, as its nearly-uniform emissions make beliefs concentrate near $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$.

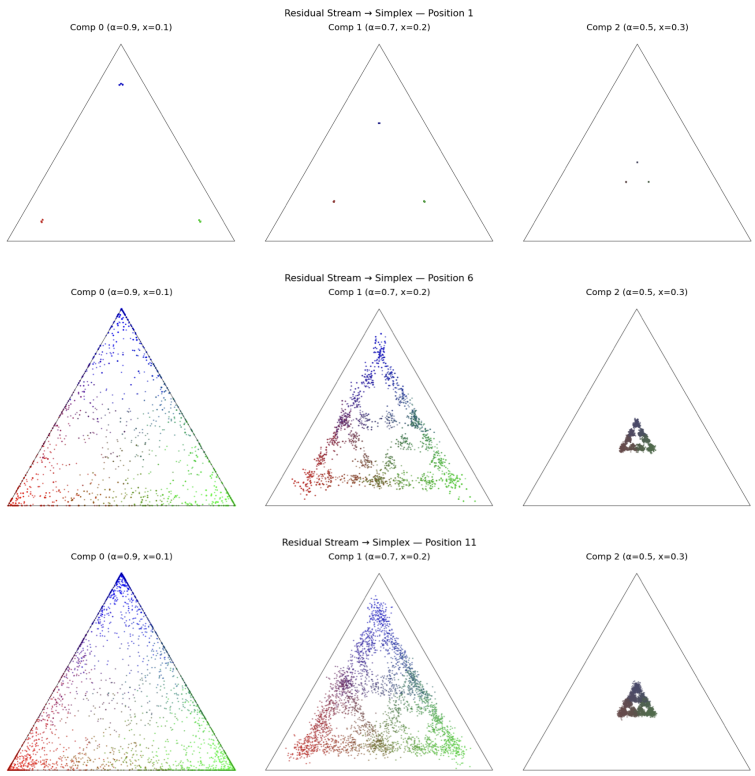


Figure 6: Residual stream projected onto the 2-simplex via per-component linear map at selected positions. Color encodes within-component belief (RGB = $P(\text{state } 0, 1, 2)$).

We also fit a single joint linear map from activations to within-component beliefs across all components simultaneously. This reveals how the three components' belief geometries relate in shared coordinates.

The fractal detail directly reflects the mixing eigenvalue $(1-3x)$: a larger eigenvalue means slower contraction toward the simplex center, so the iterated belief updates carve out more of the simplex and produce finer self-similar structure. Component 0's fractal spans most of the simplex, while Component 2's beliefs are so rapidly mixed that they never stray far from center.

The joint simplex support the shared-subspace finding from the principal angle analysis below. All three components overlap in the same region of the simplex under the shared linear map. Components 1 and 2 are concentrated in the inner region of the triangle, reflecting their faster mixing which keeps beliefs close to center. Component 0 is denser at the edges, consistent with its sharp emissions pushing beliefs toward the vertices. As context increases, these density patterns become more pronounced, but the components never separate, so the model encodes all three in the same belief subspace and distinguishes components through

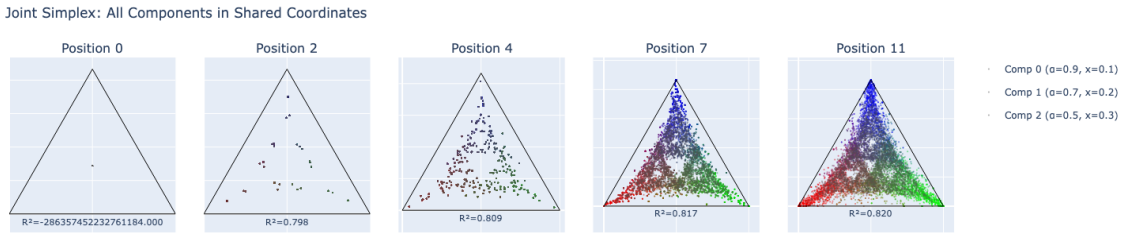


Figure 7: Joint simplex projection using a shared linear map at multiple positions. All three components projected into the same coordinates. (This is a plotly plot that can be used in the notebook where each component can be toggled.)

other dimensions.

### 4.5 Subspace Analysis

To further test whether the three components use separate or shared linear subspaces to encode beliefs, we compute principal angles between the per-component Ridge regression weight matrices. All principal angles between every pair of components are 0°:

| Pair | Principal angles |
|---|---|
| Comp 0 vs Comp 1 | $[0°, 0°, 0°]$ |
| Comp 0 vs Comp 2 | $[0°, 0°, 0°]$ |
| Comp 1 vs Comp 2 | $[0°, 0°, 0°]$ |

This means the model uses identical linear encoding directions for belief states regardless of component. This corresponds to prediction (2) from Section 2. This makes structural sense, as all three Mess3 processes have the same 3-state, 3-emission architecture and differ only in their parameters, so the belief update has the same form across components. The model appears to have learned this shared structure, reusing the same linear readout for belief tracking and encoding component identity in orthogonal dimensions.

## 5 Additional analysis 2. Attention Patterns

The model has 4 attention heads. We examine the average attention pattern for each head across all sequences, as well as the attention from the last position broken down by component.

Examining the attention patterns is informative because the mixing eigenvalues predict different optimal context windows per component: Component 0 (eigenvalue 0.7) benefits from long-range context, while Component 2 (eigenvalue 0.1) should need only the most recent tokens. If the model has learned this, we might expect component-specific attention profiles.

The heads do show some specialization. Head 1 attends along the diagonal, focusing on recent tokens. This likely functions to provide the most recent observations for belief updating. Heads 0, 2, and 3 all share similar dynamics, where they distribute attention more uniformly across the context window, gathering the longer-range statistics needed for component identification and belief refinement.

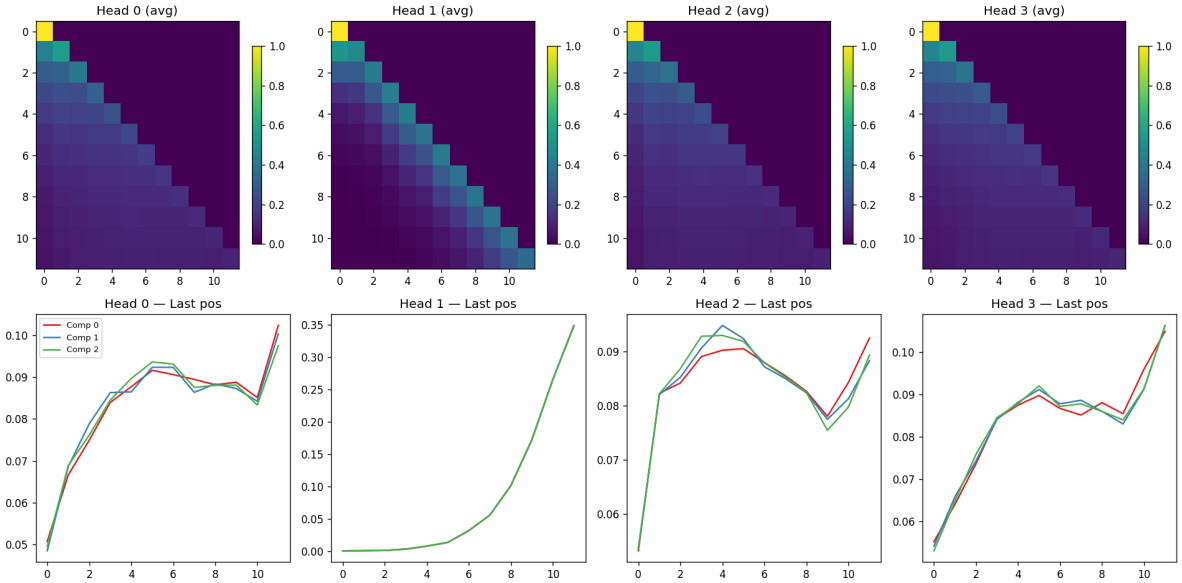


Figure 8: Top: average attention pattern per head. Bottom: attention weights from the last position to all previous positions, broken down by component.

The per-component attention profiles almost completely overlap across all four heads (and completely at head 1): the model does not modulate its attention based on which component generated the sequence. This is also consistent with the shared-subspace evidence — all components are routed through the same computational pathway.

# 6 Conclusion

My primary prediction was that the three components' geometries would superimpose early and drift apart into distinct clusters with context. Instead, evidence supports that the model implements something like prediction (2): a shared belief subspace with component identity encoded in separate dimensions.

This shared-subspace structure may connect to the effectiveness of steering vectors in large language models. If LLMs similarly separate context identity (style, persona, topic) from content processing into orthogonal subspaces, then a reasonable explanation for why steering vectors work is because they shift the representation along context-identity dimensions without disrupting the shared content machinery (e.g., a constant offset changes which "component" the model follows while leaving belief tracking intact).

There are several limitations and extensions I have in mind. Our context length of 12 means the component posterior never fully converges; with longer contexts, we could observe whether the representation collapses to a single component's fractal as the model becomes certain, producing the dimensionality decrease predicted in Section 2. It may be that the model actually wants to learn according to prediction (1), but that there is not enough context in the 12 sequence length case to get out of the shared subspace case. Additionally, testing with more components or more structurally distinct components would probe whether the shared-subspace strategy breaks down when a single belief subspace can no longer serve all components. Finally, the effective dimensionality of 5 is unexpected, and identifying what the 5th principal component encodes would sharpen our understanding of how the model structures its representations.